```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot'
)
```

```
import zipfile
import os

zip_file_path = '/content/traffic.csv.zip'
extract_dir = '/content/'
csv_file_name = 'traffic.csv' # Assuming the csv file inside the zip is name

# Unzip the file
with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
    zip_ref.extractall(extract_dir)

# Read the extracted CSV
df = pd.read_csv(os.path.join(extract_dir, csv_file_name))

df.head()
```

	event	date	country	city	artist	album	track	isrc	
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	1c
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	1c
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	34
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	08
4	click	2021-08-21	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	1c

```
df.info()
```

```
df.describe()

df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
 #   Column   Non-Null Count   Dtype
---  ------   --------------   -----
 0   event    226278 non-null  object
 1   date     226278 non-null  object
 2   country  226267 non-null  object
 3   city     226267 non-null  object
 4   artist   226241 non-null  object
 5   album    226273 non-null  object
 6   track    226273 non-null  object
 7   isrc     219157 non-null  object
 8   linkid   226278 non-null  object
dtypes: object(9)
memory usage: 15.5+ MB
```

	0
event	0
date	0
country	11
city	11
artist	37
album	5
track	5
isrc	7121
linkid	0

dtype: int64

```
df.drop_duplicates(inplace=True)
```

```
df.dropna(inplace=True)
```

```
df['date'] = pd.to_datetime(df['date'])
```

```
total_sessions = df['linkid'].nunique()
print(total_sessions)
```

```
743
```

```
pages = df.groupby('linkid')['event'].count()

bounce_rate = (pages==1).mean()*100
```

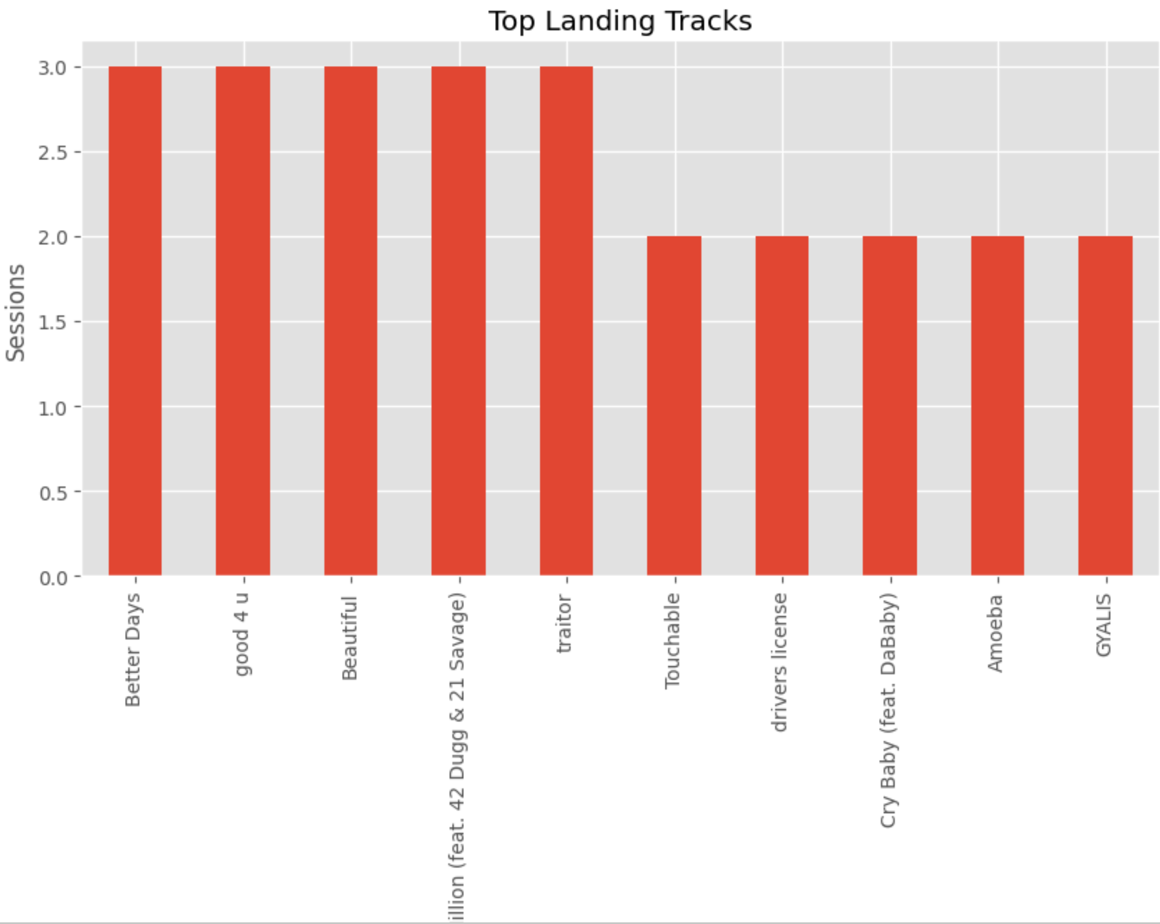
```
print(bounce_rate)
```

```
8.613728129205922
```

```
session_duration_per_linkid = df.groupby('linkid')['date'].apply(lambda x: x
avg_duration = session_duration_per_linkid.mean()

print(avg_duration)
```

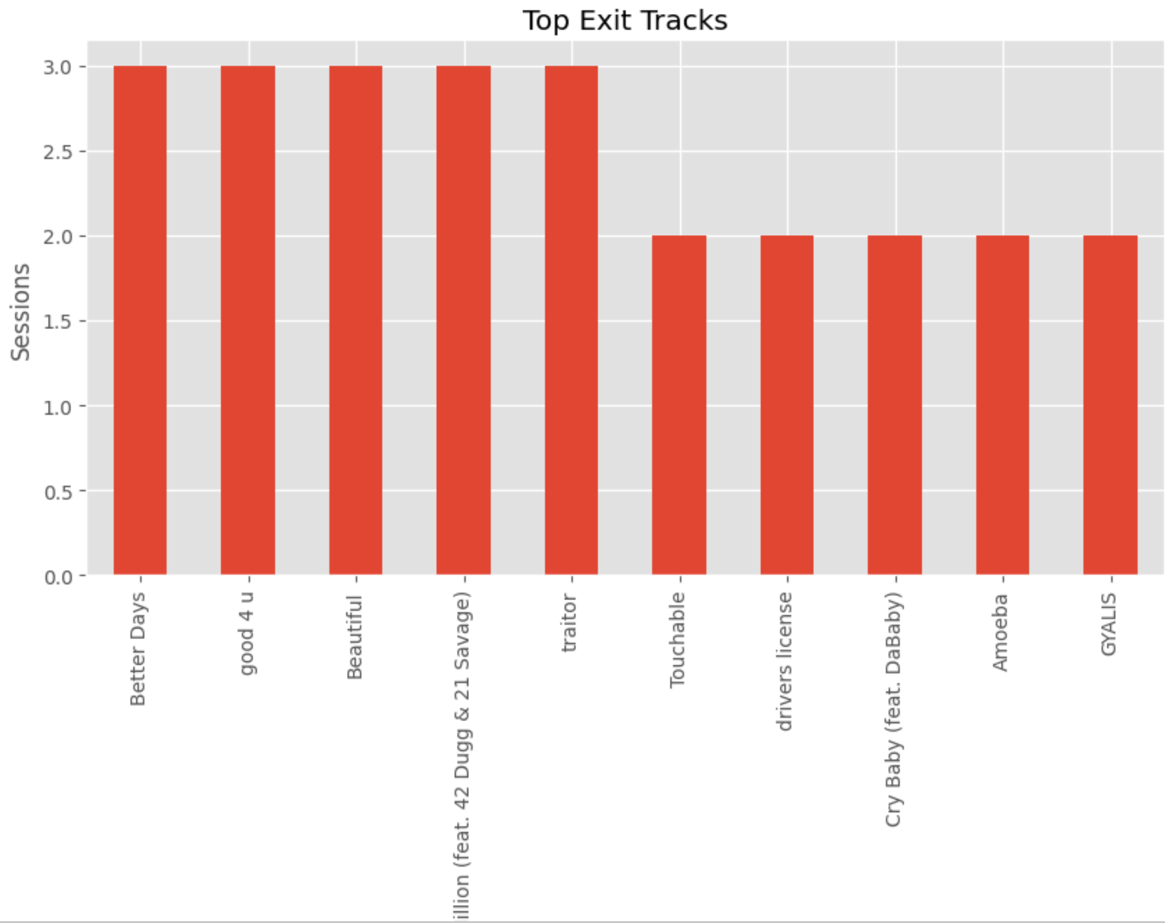
```
3 days 20:26:48.613728129
```

```
landing = df.groupby('linkid').first()

landing['track'].value_counts().head(10).plot(
kind='bar',
figsize=(10,5))

plt.title("Top Landing Tracks")
plt.xlabel("Landing Track")
plt.ylabel("Sessions")
plt.show()
```

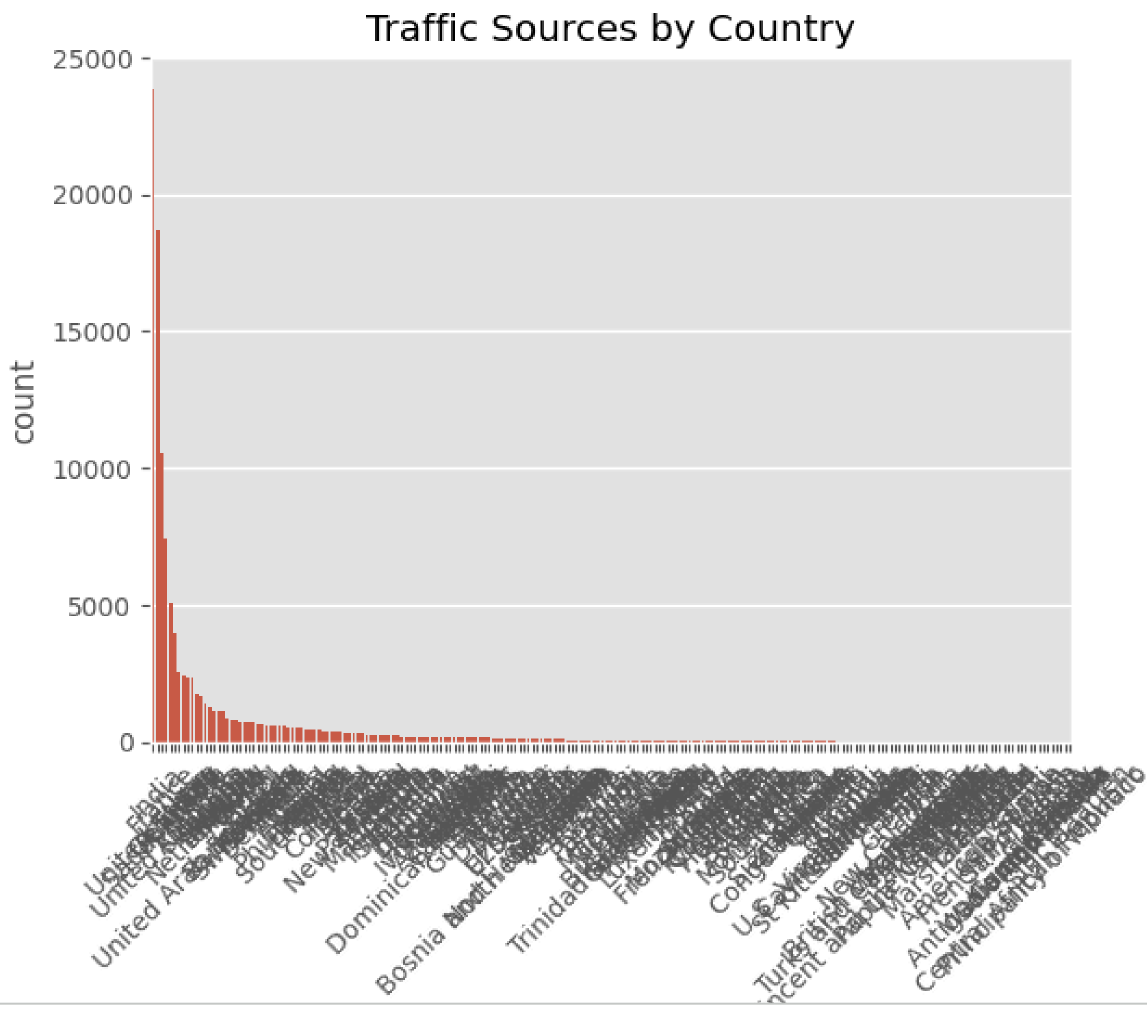



```
exit_page = df.groupby('linkid').last()

exit_page['track'].value_counts().head(10).plot(
kind='bar',
figsize=(10,5))
```

```
plt.title("Top Exit Tracks")
plt.xlabel("Exit Track")
plt.ylabel("Sessions")
plt.show()
```

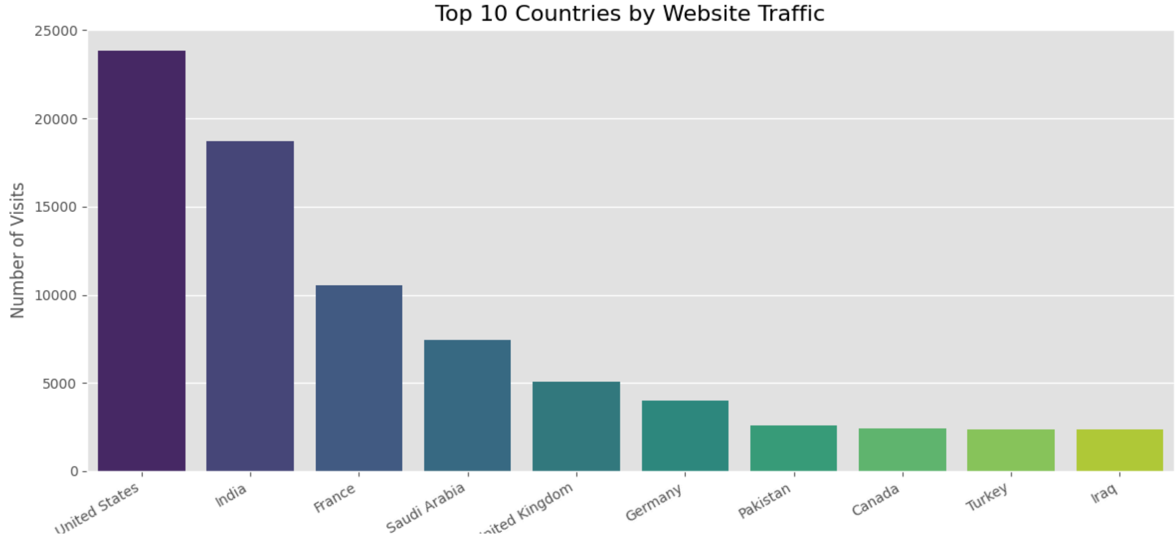



```
sns.countplot(
data=df,
x='country', # Changed 'Traffic Source' to 'country'
order=df['country'].value_counts().index)

plt.xticks(rotation=45)
plt.title("Traffic Sources by Country") # Changed title to reflect the chang
plt.show()
```

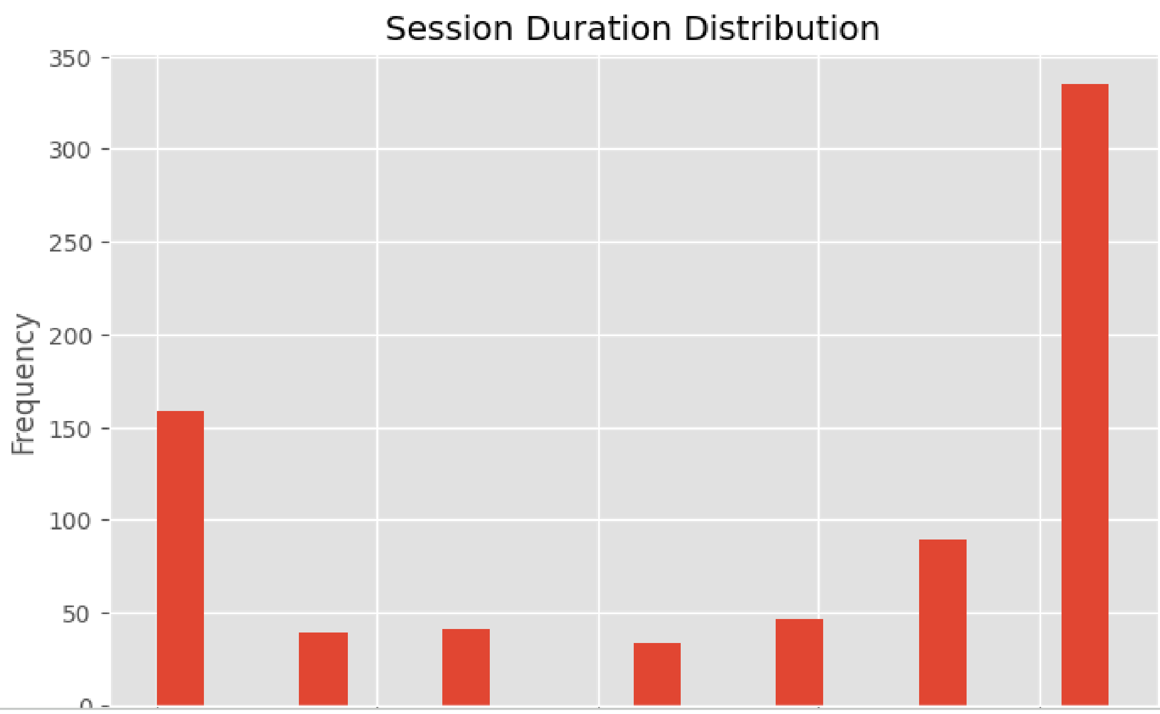



```python
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12,6))

top10 = df['country'].value_counts().head(10).index

sns.countplot(
    data=df,
    x='country',
    order=top10,
    palette='viridis'
)

plt.title("Top 10 Countries by Website Traffic", fontsize=16)
plt.xlabel("Country")
plt.ylabel("Number of Visits")
plt.xticks(rotation=30, ha='right')
plt.tight_layout()

plt.show()
```

```
/tmp/ipykernel_618/371595252.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed i

  sns.countplot(
```

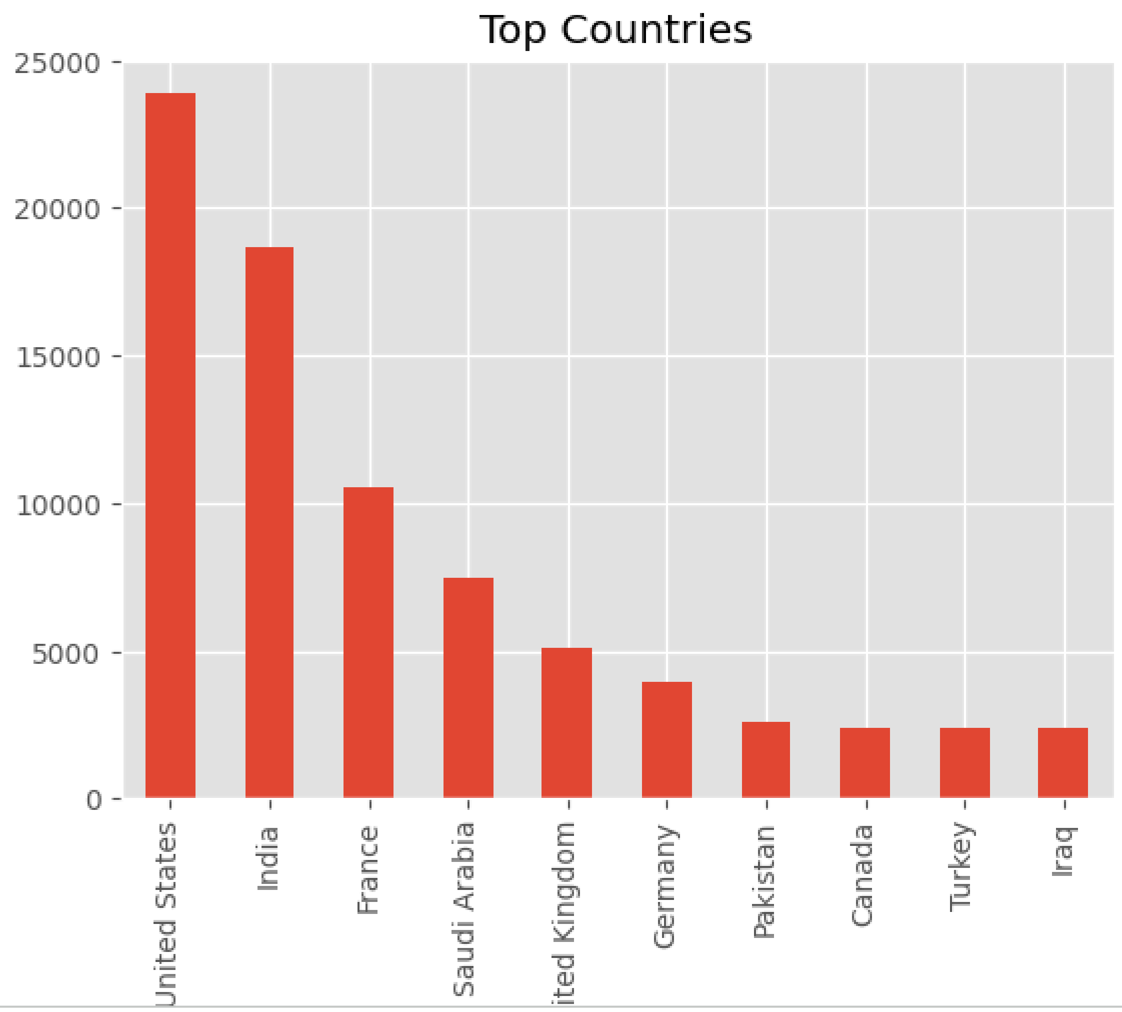



```
plt.figure(figsize=(8,5))

# Convert timedelta to minutes for plotting
plt.hist(session_duration_per_linkid.dt.total_seconds() / 60, b

plt.title("Session Duration Distribution") # Updated title for
plt.xlabel("Minutes")
plt.ylabel("Frequency")
plt.show()
```

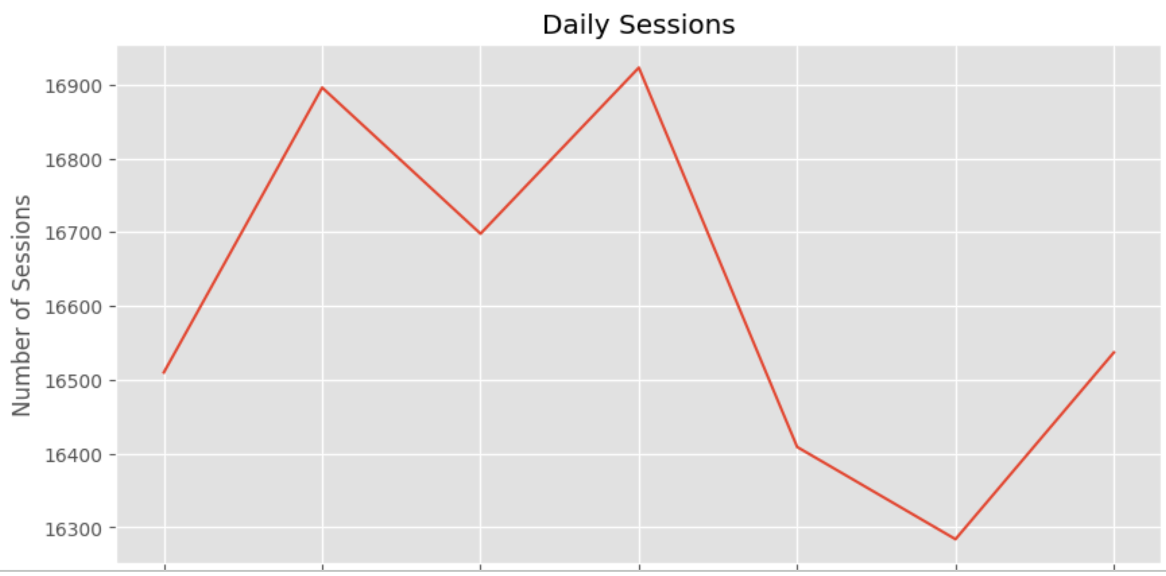



```
df['country'].value_counts().head(10).plot(
kind='bar')
```

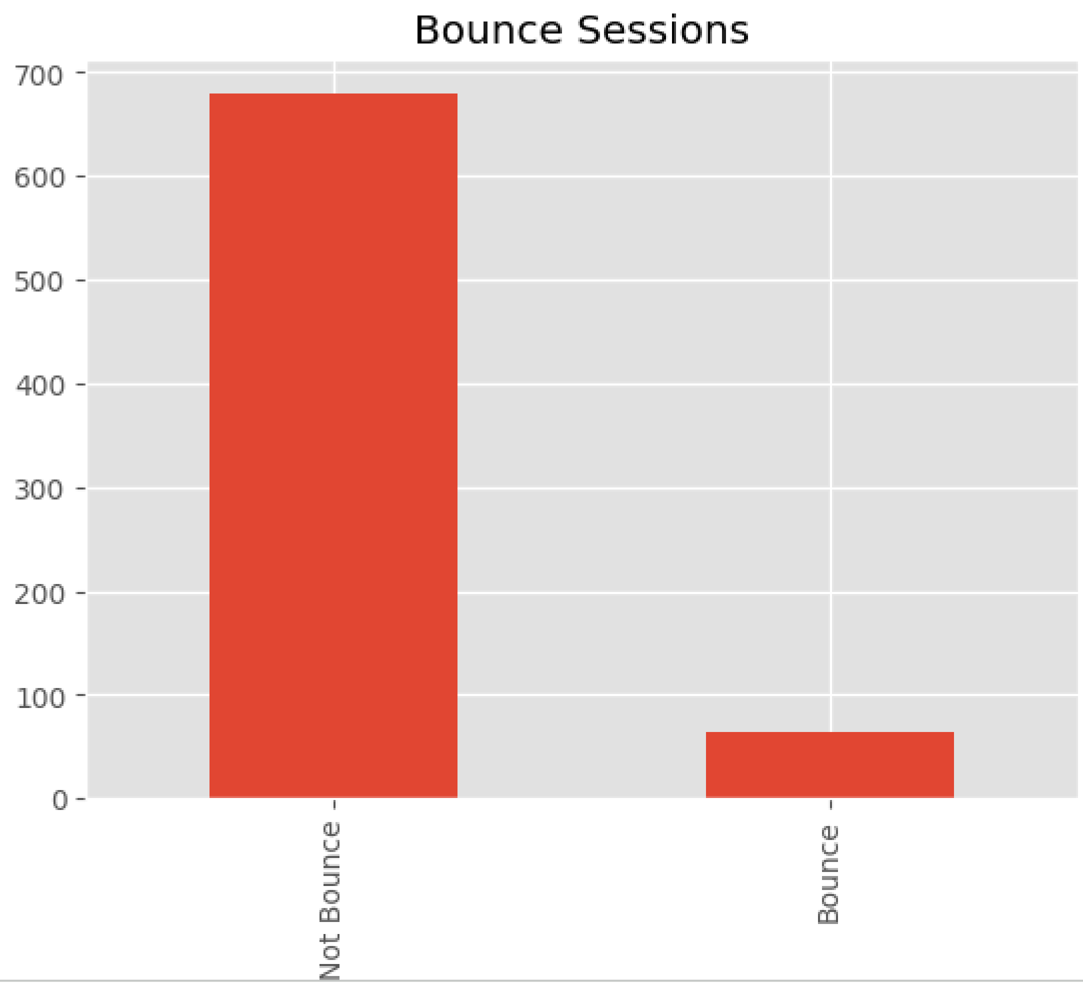
```
plt.title("Top Countries")
plt.show()
```




```
daily=df.groupby(df['date'].dt.date)['linkid'].count()

daily.plot(figsize=(10,5))

plt.title("Daily Sessions")
plt.xlabel("Date") # Added x-label for clarity
plt.ylabel("Number of Sessions") # Added y-label for clarity
plt.show()
```



```
bounce=pages==1

bounce.value_counts().plot(
kind='bar')

plt.title("Bounce Sessions")
plt.xticks([0,1],['Not Bounce','Bounce'])
plt.show()
```




```
journey=df.groupby('linkid')['track'].apply(list)

journey.head()
```

	track
linkid	
006af6a0-1f0d-4b0c-93bf-756af9071c06	[dear katara, dear katara, dear katara, dear k...
00759b81-3f04-4a61-b934-f8fb3185f4a0	[Primero Yo, Primero Yo, Primero Yo, Primero Y...
00829040-ee01-4409-966d-d67c7965144a	[Beautiful Soul, Beautiful Soul, Beautiful Sou...
009193ee-c3df-4efa-88f2-feb37c0bfdf2	[Ain't It Fun, Ain't It Fun, Ain't It Fun, Ain...
00de7566-f014-4d20-8616-82e4dea45b88	[Wiggle feat Snoop Dogg, Wiggle feat Snoop Dog...

dtype: object

User Journey

```
journey.value_counts().head(10)
```

track	count
[Deseo, ...]	
[dear katara, dear katara,	1
-	
[Primero Yo, Primero Yo, Primero Yo, Primero Yo, Primero Yo, Primero Yo,	1
[Beautiful Soul, Beautiful Soul]	1

Top Entry Pages

```
entry=df.groupby('linkid').first()

entry['track'].value_counts().head(10)
```

track	count
Better Days	3
good 4 u	3
Beautiful	3
Bout A Million (feat. 42 Dugg & 21 Savage)	3
traitor	3
Touchable	2
drivers license	2
Cry Baby (feat. DaBaby)	2
Amoeba	2
GYALIS	2

dtype: int64

Top Exit Pages

```
exit=df.groupby('linkid').last()

exit['track'].value_counts().head(10)
```

track	count
Wiggle feat Snoop Dogg]	
[Shimmy Shimmy Ya, Shimmy Shimmy Ya, Shimmy Shimmy Ya]	1
[Gordon R Freestyle, Gordon R Freestyle, Gordon R Freestyle, Gordon R Freestyle, Gordon R Freestyle]	1
[Ode To Soulja Girl (feat. AG Club)]	1
[p r i d e . i s . t h e . d e v i l (with Lil Baby), p r i d e . i s . t h e . d e v i l (with Lil Baby)]	1

dtype: int64

track	count
Better Days	3
good 4 u	3
Bout A Million (feat. 42 Dugg & 21 Savage)	3
traitor	3
Touchable	2
[illegible]	2
Cry Baby (feat. DaBaby)	2
[illegible]	2
GYALIS	2

dtype: int64

```
df.groupby('country')['linkid'].count()
```

country	linkid
Afghanistan	2
Albania	327
Algeria	515
American Samoa	4
Andorra	3
...	...
Wallis and Futuna	2
Yemen	51
Zambia	96
Zimbabwe	22
Åland	5

211 rows × 1 columns

dtype: int64